

These lecture notes are prepared by **Salman Ahmad**, who is both a **teacher** and a **web developer**. The purpose of this material is to make the topic clear and useful for students of Class 9 Computer Science.

System and Science

A system is something made of different parts that are connected and work together. Science is the way we learn and understand how those parts work and why they work like that.

Examples:

- A car 🚗 is a system, and learning how it moves is science.
- A fan 🌀 is a system, and understanding how it spins is science.
- A computer 💻 is a system, and learning how it processes data is science.
- A plant 🌱 is a system, and knowing how it grows is science.
- A human body 🧑 is a system, and learning how it works is science.
- A solar system 🌍 is a system, and understanding how planets move is science.

Natural Science

Natural science is the study of things that exist in nature (not made by humans).

👉 It helps us understand the natural world.

Examples:

- Plants 🌱
- Animals 🐮
- Weather ☁️
- Earth and space 🌍

Natural science = study of nature.

Design Science

Design science is about creating and improving things to solve problems.

👉 It focuses on man-made things.

- Building a bridge 🌉
- Making a mobile app 📱
- Designing a car 🚗
- Creating machines ⚙️

Design science = Study of human made things.

Computer Science

Definition: Computer science studies how computers work, what they can do, and their limitations. It uses both natural science and design science methods.

Natural Science in Computer Science

Focuses on understanding basic rules of computer systems, how they work.

Study Of Algorithm Studying algorithms to see how fast or efficient they are. QuickSort and MergeSort are sorting algorithms; researchers analyze how they sort data efficiently.

Design Science in Computer Science

Design science in computer science means using computers to design and create useful systems, programs, and solutions that solve real-life problems.

Examples:

- Development of new software tools: It means creating new computer programs or applications that help people do tasks easily and solve problems.

Examples:

- A new mobile app for learning
-  A photo editing tool 
- A game application 
- A banking software system 
- Improvement of Computer Systems: It means making a computer system faster, better, safer, and more useful by upgrading hardware or software.

Examples:

- Updating Windows or mobile system 
- Adding more RAM to a computer
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- Installing antivirus for safety 
- Upgrading software to a new version 

Computer as a System

Definition: A computer is a complex system designed to process data, perform tasks, and follow instructions efficiently.

Objective: Perform computations, process information, and run software applications like word processors, web browsers, and games.

Components of a Computer

1. Interface Components

Parts that allow the user to interact with the computer.

Examples: Keyboard, mouse, monitor, printer.

2. Processing Components

- CPU (Central Processing Unit): Often called the “brain” of the computer. It performs all calculations, executes instructions, and controls the overall operation of the system by coordinating with other components.
- RAM (Random Access Memory): Temporary storage for data and instructions being used by the CPU.
- Storage (Hard Drive/SSD): Long-term storage for programs and data. It keeps information saved even when the computer is powered off, allowing access whenever needed.

Note: HDD Stands for Hard Disk Drive and SSG stands for solid State Drive. [SSD is 10 times faster and 10 times expensive than HDD.]

- Operating System: Manages computer operations and communicates with hardware.
- Application Software: Programs that perform specific tasks, like Microsoft Word or Photoshop.

3. Communication Components

- Motherboard: Connects all parts of the computer.
- System Bus: Transmits data between CPU and other components.
- Types of Buses:
 - Data Bus: Transfers actual data between the CPU, memory, and other components.
 - Address Bus: Carries memory addresses that specify where data is stored or should be accessed.
 - Control Bus: Sends control signals to coordinate and manage operations within the computer system.

Interaction Among Components

Example: Opening a file ("report.docx")

- ◊ User Action: Double-click the file or press a key combination.
- ◊ Input Device: Mouse or keyboard sends the signal to the CPU.
- ◊ Processing: CPU retrieves the file from storage, using RAM for temporary operations.
- ◊ Output Device: File opens on the monitor.
- ◊ Power Supply: Provides electricity for the computer.
- ◊ Network: Connects the computer to other systems or the Internet.
- ◊ Peripherals: Devices like printers, scanners, and external drives.